DHINESH SWAMINATHAN M

Email Id: dhineshswaminathan03@gmail.com
Phone No: 9751735525
Linkedin: www.linkedin.com/in/dhineshswaminathan
Portfolio: https://dhineshswaminathan.vercel.app/

PROFESSIONAL SUMMARY

MCA graduate with a strong foundation in full-stack development, building scalable web applications using React.js, FastAPI, Java, Python, and SQL. Experienced in designing REST APIs, optimizing application performance, and developing AI-powered solutions. Passionate about building reliable software products and solving real-world engineering challenges.

TECHNICAL SKILLS

Programming Languages: Java, C, JavaScript, TypeScript

Web Technologies: React.js, HTML, CSS, Tailwind CSS

Tools & Frameworks: Python FastAPI, Vite, Git, GitHub, React Query

Database: MySQL, PostgreSQL, Supabase

EDUCATION

Master of Computer Applications (MCA) (2024-2026)
P.S.N.A College of Engineering and Technology | CGPA: 8.6
Bachelor of Science in Mathematics (2021 – 2024)
Thiagarajar College, Madurai | CGPA: 8.53

PROJECTS

Real-Time Computer Vision Surveillance Platform

- Developed a full-stack edge AI surveillance platform using React.js, FastAPI, OpenCV, and YOLOv8 for real-time weapon, fight, unattended baggage, and crowd detection.
- Engineered a multi-model computer vision pipeline by combining custom YOLOv8 object detection with pose estimation for real-time threat detection.
- Built a FastAPI backend with asynchronous processing and REST APIs for low-latency video analysis and frontend integration.
- Designed a React.js dashboard with secure routing, state management, and HTML5 Canvas overlays for live alerts and bounding-box visualization

Market Buddy – Full-Stack Grocery Store Platform

- Built a responsive React frontend for product browsing, cart management, and checkout
- Developed secure admin panel with full CRUD operations and real-time inventory/order updates via Supabase
- Managed complex client-side state with React Hooks; enforced TypeScript schemas for frontend-backend consistency
- Delivered high-performance, accessible UI using Tailwind CSS and shadcn-ui

Cloud-Native Micro File Sharing Platform

- Developed secure file-sharing system handling 100+ requests using FastAPI and React.js
- Improved API response time by ~25% via SQL query optimization
- Implemented access-based authentication and resolved backend issues

Research & Publications

AI-Driven Real-Time Crowd Analytics and Anomaly Detection for Public Safety in High-Density Environments

Presented at International Conference on Contemporary Mathematics and Innovations (ICCMI-2026),

Thiagarajar College of Engineering, Madurai|July 2026

CERTIFICATIONS

C++, C, Advanced C++ Spoken Tutorial — IIT Bombay

LaTeX Certification (98%)